

Lab 9 Development Report

Application and Deployment Enhancement

CS 5542 — Big Data and Analytics
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1. PROJECT OVERVIEW

Our team has built and iteratively improved an AI-powered logistics platform across Labs 5–8 and integrated all components into the Phase 2 project: HyperLogistics, a Snowflake-native neuro-symbolic supply chain resilience system. Lab 6 delivered a Multi-Agent Trucking Analytics Dashboard. Lab 7 reproduced and extended ReMindRAG. Lab 8 introduced the Consensus Planning Protocol (CPP) with a domain-adapted Phi-2 safety validation agent. Lab 9 directly addresses instructor feedback from Labs 6 and 7, fixes critical Lab 8 evaluation failures, and integrates all three components into the unified Phase 2 system for Research-A-Thon demonstration.

Phase 2 GitHub: https://github.com/mosomo82/CS5542_SmartSC_Optimization_System

Phase 2 Live App: <https://cs5542hyperlogistics.streamlit.app/>

Lab 6 Live App: <https://cs5542logisticsai.streamlit.app/>

Lab 8 Live App: <https://cs5542lab8.streamlit.app/>

Lab 6–9 Repo: https://github.com/mosomo82/COMP_SCI_5542

Lab 7 Repo: https://github.com/mosomo82/ReMindRAG_Week7

2. PRIOR LAB FEEDBACK ADDRESSED

Lab 6 Feedback

- *"Minor improvements could include expanding the documentation of system architecture and evaluation details."*

Response: ARCHITECTURE.md and EVALUATION.md authored for Week_9/. Section 3.5 below.

Lab 7 Feedback

- *"Minor improvements could include expanding the benchmarking results and adding additional automated verification for reproducibility experiments."*

Response: BENCHMARKING.md and REPRO_VERIFICATION.md authored (suite expanded 17→26 tests with CI). Section 3.6 below.

Lab 8 — Proactive Fixes

Three critical issues identified in the Lab 8 evaluation are resolved in Lab 9: the Q13→Q14 monotonicity failure (CPP Spatial SQL hard gate), hallucinated bridge constraints (evidence injection), and a contradictory accuracy number in the report (corrected). Section 3.7 below.

3. APPLICATION ENHANCEMENTS (LAB 9)

3.1 Lab 6 — UI and Stability

- Persistent sidebar, Agent Chat tool-call trace expander, Reset Session button, SQL Explorer error UX.

- Snowflake keep-alive ping: cold-start latency ~8s → <2s. Exponential-backoff retry on all Gemini API calls.
- Python logging module replacing print statements. FastAPI /health endpoint + UptimeRobot uptime monitoring.
- Agent eval suite expanded 5→10 scenarios with GitHub Actions CI (≥70% pass-rate gate).

3.2 Lab 7 — Benchmarking and Verification

- BENCHMARKING.md: constrained LooGLE run (F1=0.496, consistent with paper's 0.49), HotpotQA subset (F1=0.58 vs paper's 0.61), efficiency verification (37–42% API call reduction on repeated queries).
- REPRO_VERIFICATION.md: 9 new automated tests across test_benchmark_smoke.py and test_repro_variance.py. Suite expanded 17→26 tests at zero API cost, fully integrated into CI.

3.3 Lab 8 — Critical Fixes

- CPP Spatial SQL hard gate: bridge constraint check before any LLM invocation. Monotonicity invariant (APPROVE→VETO on bridge violation) is now deterministic SQL — all 13 metamorphic tests pass.
- Evidence injection: [RETRIEVED CONSTRAINTS] block injected verbatim into prompt. Model instructed not to recall limits from memory. Hallucination rate drops from ~40% to 8%.
- Accuracy discrepancy corrected: baseline was 40% on both platforms (not 60% as stated in Section 5 intro). EVALUATION.md documents correct values.
- CoT-formatted training data: generate_dataset.py updated for full 4-step chains. Dataset expanded 100→300+ pairs across 5 disruption types. PEFT CoT quality: 67%→83%.
- Evaluation expanded 15→50 queries with 5-dimension rubric (0–10). Metamorphic suite expanded 3→13 pairs. GitHub Actions CI added.

3.4 Lab 6 Documentation

- ARCHITECTURE.md (Week_9/): 4-layer system, agent lifecycle, 9 tools, observability stack.
- EVALUATION.md (Week_9/): 10-scenario methodology, S2 root-cause, CI integration, traceability table.

3.5 Lab 7 Documentation

- BENCHMARKING.md: paper-vs-repro comparison, constrained runs, variance table, JSON output schema.
- REPRO_VERIFICATION.md: 26-test suite design, CI YAML, manual verification table.

3.6 Phase 2 Integration (CS5542_SmartSC_Optimization_System)

All Lab 6–8 components are integrated into the Phase 2 unified system for Research-A-Thon demonstration:

- CPP agent pipeline (src/agents/): compliance_agent.py implements the Spatial SQL hard gate; context_agent.py wires ReMindRAG into the retrieval layer; evidence injection is applied at the prompt level.
- Unified Streamlit dashboard (src/app/dashboard.py): merges Lab 6 analytics tabs (fleet, routes, safety, fuel) with the Lab 8 CPP dispatcher interface. Persistent sidebar, reasoning path expander, and retry decorator applied throughout.
- Data pipeline (src/run_pipeline.py): logging module replaces print statements; verify_pipeline.py adds row-count and schema assertions for all 4 SILVER tables.
- CI/CD (.github/workflows/ci.yml): pipeline smoke test + CPP unit tests + system eval (mock mode) + ReMindRAG 26-test suite — all run automatically on every push with a ≥70% pass-rate gate.
- ARCHITECTURE.md + EVALUATION.md at Phase 2 repo root: full system-level reference documentation covering the 4-layer architecture, CPP detail, RAG benchmarks, and pipeline evaluation

results.

4. EXTENSION OF PHASE-2 PROTOTYPE

Lab	Capability	Baseline	Lab 9 Enhancement
6	Architecture docs	README only	ARCHITECTURE.md + EVALUATION.md (Week_9/)
6	UI & monitoring	9 raw tabs, manual eval	Sidebar, tool trace, 10 scen., CI
6	Logging & stability	Print stmts, ~8s cold-start	Logging, retry, <2s, /health
7	Benchmarking	Metrics unverified	LooGLE F1=0.496, HotpotQA F1=0.58
7	Repro. verification	17 structural tests	26 tests + CI (zero API cost)
8	Monotonicity failure	FAIL both platforms	PASS — Spatial SQL hard gate
8	Hallucination (PEFT)	~40% rate	8% — evidence injection
8	Evaluation coverage	15 queries, binary scorer	50 queries, 5-dim rubric (0–10)
P2	CPP integration	Architecture described only	compliance_agent.py + wired CPP
P2	Unified dashboard	Separate lab apps	Single app: CPP + Lab 6 analytics
P2	System CI	None	GitHub Actions: pipeline+eval+RAG
P2	System docs	README only	ARCHITECTURE.md + EVALUATION.md (root)

5. REPOSITORY AND DEPLOYMENT LINKS

Resource	URL
Phase 2 Live App	https://cs5542hyperlogistics.streamlit.app/
Phase 2 GitHub	https://github.com/mosomo82/CS5542_SmartSC_Optimization_System
Lab 6 Live App	https://cs5542logisticsai.streamlit.app/
Lab 6–9 Main Repo	https://github.com/mosomo82/COMP_SCI_5542
Lab 7 ReMindRAG Repo	https://github.com/mosomo82/ReMindRAG_Week7
Phase 2 ARCHITECTURE.md	github.com/.../CS5542_SmartSC.../blob/main/ARCHITECTURE.md
Phase 2 EVALUATION.md	github.com/.../CS5542_SmartSC.../blob/main/EVALUATION.md
Lab 8 ARCHITECTURE.md	github.com/.../COMP_SCI_5542/blob/main/Week_8/ARCHITECTURE.md
Lab 7 BENCHMARKING.md	github.com/mosomo82/ReMindRAG_Week7/blob/main/BENCHMARKING.md

6. SUMMARY

Lab 9 closes all identified gaps across Labs 6, 7, and 8 while completing the Phase 2 integration of HyperLogistics as a unified, demonstrable system. Lab 6 and 7 instructor feedback is addressed by six new documentation artifacts. Lab 8's three critical failures are resolved by deterministic SQL, evidence injection, and expanded evaluation. The Phase 2 repo gains a fully wired CPP pipeline, a unified Streamlit dashboard, end-to-end CI, and system-level architecture and evaluation documentation — positioning HyperLogistics for

the Research-A-Thon competition.